

🧠 Task - JavaScript (ES6 + Loops + Higher Order Functions)

Part 1 - Choose

(1) إيه اللي بيرجعه `map`() ؟

- [] أول عنصر يحقق شرط

- [] Array جديدة بنفس الطول

- [] Boolean

- [] Number

(2) مين فيهم بيرجع أول عنصر يحقق الشرط؟

- [] filter()

- [] map()

- [] find()

- [] forEach()

... بيرجع `filter`() 3) ###

- [] أول عنصر

- [] Array جديدة بالعناصر اللي حققت الشرط

- [] Number

- [] String

... بيرجع `forEach`() 4) ###

- [] Array جديدة

- [] أول عنصر

- [] undefined

- [] Boolean

... بنستخدمها غالباً مع `for...of` 5) ###

- [] Objects

- [] Arrays
- [] Functions
- [] Classes

Part 2 - True or False

1. `map()` الأصلية Array يغير الـ **false**
2. `filter()` فاضية Array ممكن يرجع **true**
3. `find()` ممكن يرجع undefined **true**
4. `for...in` Array بتاع الـ Index بيلف على الـ **true**
5. `forEach()` break ينفع أعمل بيها **false**

Part 3 - Complete the following

Q1

خلي الكود يطبع:

```
...  
2  
4  
6  
8  
...
```

```
```js  
const numbers = [1,2,3,4];

numbers.map((num)=>{
 console.log(num * 2);
});
```
```

Q2

طلع Array فيها الأرقام الأكبر من 20.

```

``js
const nums = [10,25,5,30,15,40];

const result = nums.filter((num)=>{
  return num > 20;
});

console.log(result);

```

Q3

هات أول شخص عمره أكبر من 25.

```

``js
const users = [
  {name:"Ali", age:20},
  {name:"Sara", age:28},
  {name:"Omar", age:30}
];

const user = users.find((item)=>{
  return item.age > 25;
});

console.log(user);

```

Q4

حوّل كل الأسماء لـ Uppercase.

```

``js
const names = ["ali","mona","ahmed"];

const result = names.map((name)=>{
  return name.toUpperCase();
});

```

```
console.log(result);  
...
```

Part 4 - To Do

عندك الـ Array دي

```
``js  
const fruits = ["Apple","Banana","Orange"];  
...
```

1)

اطبع كل عنصر باستخدام `for...of`

```
const fruits = ["Apple", "Banana", "Orange"];  
let item = "";  
for (let x of fruits) {  
    item += `${x}\n`;  
}  
console.log(item)
```

2)

اطبع الـ Index باستخدام `for...in`

```
const fruits = ["Apple", "Banana", "Orange"];  
for (let item in fruits) {  
    console.log(item)  
}
```

3)

اطبع بالشكل ده باستخدام `forEach`

...

0 -> Apple

1 -> Banana

2 -> Orange

...

```
const fruits = ["Apple", "Banana", "Orange"];

fruits.forEach(function (fruit, index) {
    console.log(`${index} -> ${fruit}`);
});
```

Part 5 - To Do

Q1

حوّل الكود لـ Arrow Function

```
``js
function sum(a,b){
    return a+b;
}
...

```

```
const sum = (a, b) => a + b;
```

Q2

استخدم Destructuring

```
``js
const user = {
    name:"Mostafa",
    age:25
};
...

```

خد `name` و `age` في متغيرات.

```
const user = {  
  name: "Mostafa",  
  age: 25  
};  
  
const { name, age } = user;  
  
console.log(name);  
console.log(age);
```

Q3

استخدم Template Literal

بدل

```
``js  
console.log("Hello " + name);  
``
```

```
console.log(`Hello ${name}`);
```

Q4

استخدم Spread Operator

```
``js  
const arr1 = [1,2,3];  
const arr2 = [4,5,6];  
``
```

اعمل Array واحدة فيها الكل.

```
const arr1 = [1, 2, 3];
const arr2 = [4, 5, 6];

const allNumbers = [...arr1, ...arr2];

console.log(allNumbers);
```

Part 6 - Many Q

عندك البيانات دي:

```
``js
const students = [
  {name:"Ali", degree:70},
  {name:"Sara", degree:95},
  {name:"Ahmed", degree:40},
  {name:"Mona", degree:85},
  {name:"Omar", degree:55}
];
...

```

Required

1)

اعمل Array فيها أسماء الطلبة بس.

```
const students = [
  {name:"Ali", degree:70},
  {name:"Sara", degree:95},
  {name:"Ahmed", degree:40},
  {name:"Mona", degree:85},
  {name:"Omar", degree:55}
];

const names = students.map((student) => student.name);
```

```
console.log(names);
```

2)

اعمل Array فيها الطلبة اللي درجاتهم أكبر من أو تساوي 60.

```
const students = [
  {name:"Ali", degree:70},
  {name:"Sara", degree:95},
  {name:"Ahmed", degree:40},
  {name:"Mona", degree:85},
  {name:"Omar", degree:55}
];

const passedStudents = students.filter((student) => student.degree >= 60);

console.log(passedStudents);
```

3)

هات أول طالب درجته أكبر من 90.

```
const students = [
  {name:"Ali", degree:70},
  {name:"Sara", degree:95},
  {name:"Ahmed", degree:40},
  {name:"Mona", degree:85},
  {name:"Omar", degree:55}
];

const topStudent = students.find((student) => student.degree > 90);

console.log(topStudent);
```


4)

اطبع أسماء كل الطلبة باستخدام `forEach().`

```
const students = [
  {name:"Ali", degree:70},
  {name:"Sara", degree:95},
  {name:"Ahmed", degree:40},
  {name:"Mona", degree:85},
  {name:"Omar", degree:55}
];

students.forEach((student) => {
  console.log(student.name);
});
```

Bonus

بدون استخدام Loop عادية (`while` أو `for`)

احسب مجموع الأرقام دي باستخدام `reduce()` لو فاكرها أو دور عليها في الـ MDN

```
```js
const numbers = [5,10,15,20];
```
```

```
const numbers = [5, 10, 15, 20];

const sum = numbers.reduce((total, number) => total + number, 0);

console.log(sum);
```